Summary of federated learning research paper

Luo Xiaoyu – 50087474

This paper, "Communication-Efficient Learning of Deep Networks from Decentralized Data," proposes a novel neural network training model: Federated Learning (FL). This learning paradigm is primarily used in modern mobile devices (such as smartphones and tablets) to train machine learning models using data generated from these dispersed devices. These devices often generate a wealth of compelling data, such as personal health metrics, SMS history, or geolocation logs—data that is typically valuable and highly sensitive to privacy. Furthermore, due to the fragmented and massive amount of data, it's impossible to record it in a centralized data center for unified training as in traditional training. The concept of Federated Learning arose to address this, its core idea being to separate the model's training capability from the need to store raw training data in a single location. This not only protects user data privacy but also alleviates data transmission problems in unreliable network environments.

1. Client-Server Architecture: The basic architecture of Federated Learning is the Client-Server Architecture. The entire training process revolves around the interaction between a central server and multiple distributed clients.

   Central Server: Responsible for maintaining the global model and coordinating the entire training process. It initializes the global model and distributes it to all available clients. Once a client returns its trained model, the central server performs a weighted average of the parameters from the client's local model based on the data, resulting in the final new global model.

   Client: Responsible for training its local model. It receives the global model from the central server and uses it to train its local data (often sensitive data). After training, it uploads the updated local model back to the central server. Sensitive local data remains stored on the client and is protected from intrusion.

2. Data Privacy: Data privacy is one of the main reasons for the emergence of federated learning. In traditional machine learning, all data is centrally stored in a data center for training, which can lead to the potential for the exploitation of sensitive data. Federated learning avoids this by implementing the principle of "data minimization." Data remains on distributed clients; only the trained model parameters are sent back to the central server. This approach not only prevents the malicious use of sensitive data but also decouples data from model training, facilitating the inclusion of more clients in the training process.

3. Non-IID Data and Communication Cost: However, the privacy protection inherent in federated learning also presents certain challenges, primarily two:

   Non-IID Data: Data between different clients is typically highly independent, and the data on different users' phones generally varies significantly. This data distribution characteristic leads to strong "bias" in the models trained on each client, complicating model convergence and even drastically reducing accuracy.

Communication Cost: Another drawback of federated learning models is the communication cost. Distributed clients, especially mobile devices, rely on internet connections that are typically slow, unstable, and expensive. Achieving communication for model training in this environment is a significant challenge.

In summary, the main contribution of this paper is the proposal of the Federated Averaging (FedAvg) algorithm, a fundamental framework for federated learning. FedAvg achieves communication efficiency by introducing a key parameter: allowing clients to perform multiple epochs ($E > 1$) of SGD training using local data before each communication, with gradient updates performed in mini-batches (batch size = B) per epoch. The essence of this mechanism is to increase local computation to reduce communication costs. Various experiments in the paper show that FedAvg is highly effective; compared to synchronous SGD, the algorithm reduces the number of communication rounds by 10 to 100 times while achieving the same model accuracy.